Eclipse: simulating election manipulation in multi-agent networks

Simulator, configuration, studies, results and certified equilibria · state at commit 86de48b, 9 November 2026

splittersplatter/Multi_Agent_System

At a glance: one engine, seven mechanisms, every run reproducible

343
tests passing (re-run, 35 s)

26
built-in voting rules + plugins

7
manipulation mechanisms + 17-arm matrix

5
certified convergence domains (F1–F5)

0
failed runs across all re-run studies

1 What works

Restricted strategic play converges to provable equilibria (F1–F5). Budgeted coalition sight and eavesdropping, campaign edges and alignment attacks each flip small elections in 5 of 5 seeds.

2 What doesn't

There is no universal convergence guarantee: a two-voter conformity cycle refutes it. Distorted polls with a budget of 2 and hidden leader adaptation change no winners in the example electorates.

3 What is next

Stage 11 needs a frozen protocol: balanced electorates, at least 30 held-out seeds per scenario and run-level uncertainty. Example-scale results validate the controls; they are not effect sizes.

Research question: how far can bounded manipulators move an election?

Voters see only local, possibly corrupted information. Every manipulator spends from an explicit, logged budget.

1 Dynamic proximity

Contacts change as voters move; compared against frozen and static graphs

2 Coalitions

Preference sight, deceptive reports, private-channel eavesdropping

3 Campaign edges

Buy directed influence edges from supporters to non-supporters

4 Public-poll distortion

Bounded L1 shifts of a broadcast poll toward a target

5 Spatial nomination

Leaders nominated from issue-space proximity

6 Alignment attacks

Push voter ideals toward a target, with an aggregate-only detector

7 Leader adaptation

Hidden platform moves revealed only through noisy signals

Combined system

All seven together, with isolated and leave-one-out ablations

Roadmap: twelve cumulative stages on one simulator

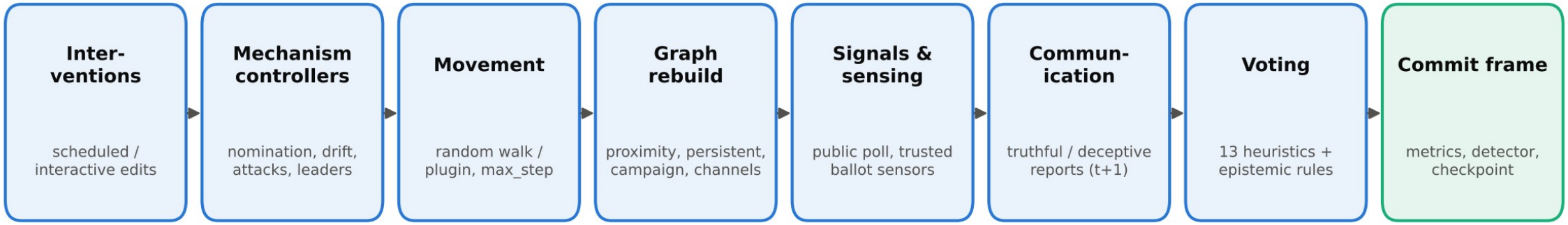
00 Baseline & research contract Verified	**01** Simulator, CLI, browser, replay Implemented	**02** Epistemic voting Implemented	**03** Dynamic proximity Implemented	**04** Coalitions Implemented	**05** Campaign edges Implemented
06 Public-poll distortion Implemented	**07** Nomination & alignment Implemented	**08** Alignment attacks & detector Implemented	**09** Adapting leaders Implemented	**10** Combined system Implemented	**11** Frozen experiments & report Pending

Built 5–9 November 2026 across 7 commits. The regression suite grew from 130 to 343 tests as stages landed.

“Implemented” means the code runs and is tested. The formal stage acceptance gates (manual browser checks, rejected-action audit) are still open.

One round, eight phases, one atomic commit

One simulation round (engine.step): every phase reads the previous committed frame, then commits atomically



Rewind/edit replaces the future · keyed SHA-256 RNG per actor/round/phase · replay verifies every checkpoint

Separated state

Private preferences, cast ballots, reports and perceptions are never merged.

Delayed information

A report generated at round t describes round t−1 and is delivered at t+1.

Paired randomness

SHA-256-keyed RNG per actor, round and phase keeps treatment and control arms matched.

Architecture: one engine behind every interface

Configuration model.py · canonical TOML parsing, immutable schemas

Coordinator engine.py · phase order, atomic frames, rewind/resume

Observations observations · coalitions · policies (scoped inputs)

Decisions voting · epistemic · network · campaign · signals · detector · majority

Replay records · provenance: SHA-256 of core, pipeline, web, plugins

Evaluation metrics · *_experiments · combined · majority_experiments

CLI

18 commands: run, resume, batch, compare-*, export, knowledge

Browser

Play, step, rewind, edit, inspect; network and statistics views

Exports

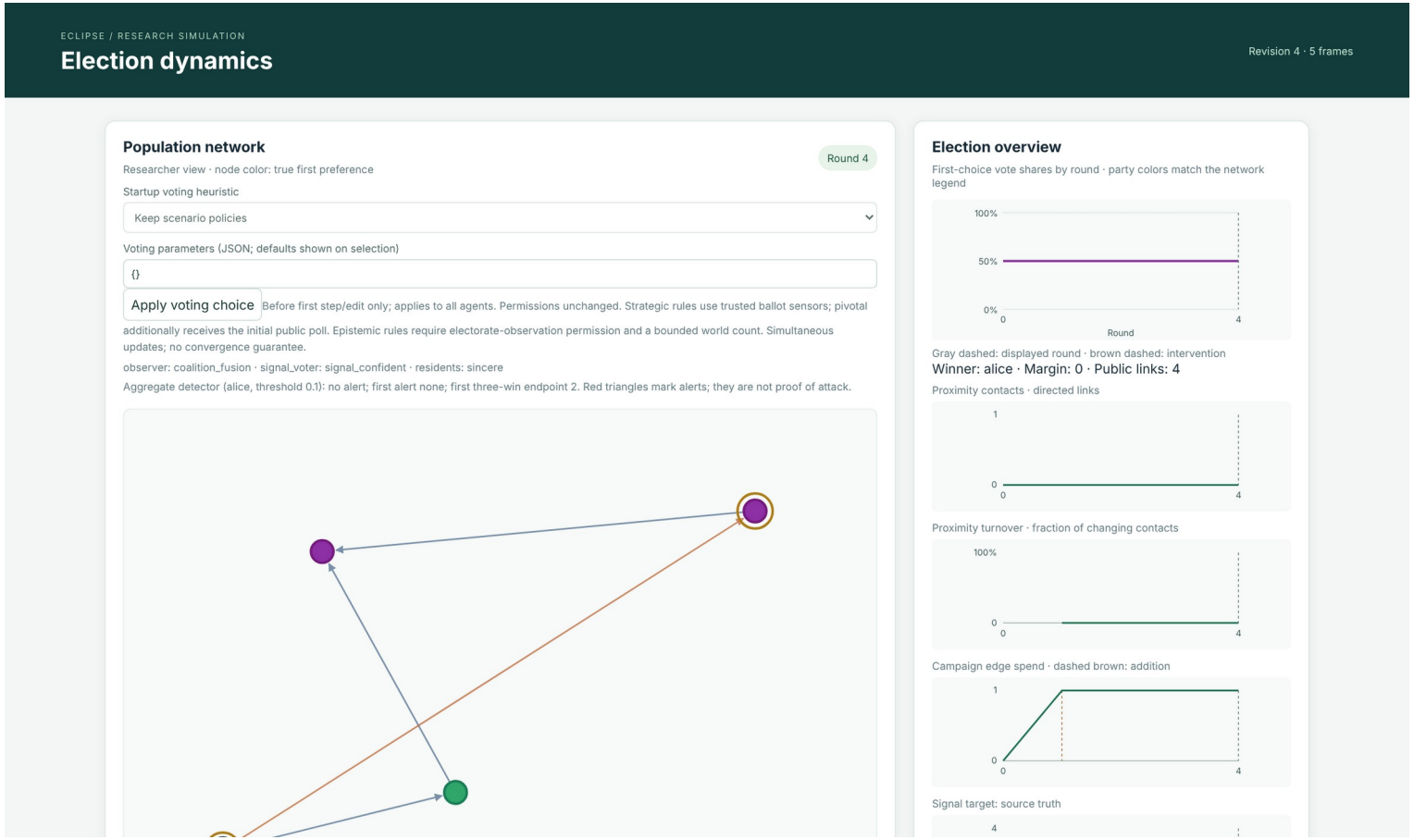
CSV, SVG, PNG and GIF from recorded frames

Checkpoints

v21/v22, verified by full replay; tampering rejected

6.0k lines of engine Python, 0.5k of web UI, 6.9k of tests. Core uses only the standard library; Pillow is optional for images.

Interactive browser: watch, rewind and edit an election



Population network: persistent (blue), proximity (grey) and purchased campaign (orange) edges

Charts: vote shares, contacts, turnover, campaign spend, signals, detector

Inspector: each agent's permissions, trusted observations, reports and policy memory

Edits: editing a past round replaces the future; drafts are per agent and round; stale revisions are rejected

Startup heuristic selector switches all 13 voting rules before round 1

Configuration I: a scenario is one TOML file

Key / table	Definition and bounds
`parties, k`	Candidate IDs (also the tie order); global ballot length $1 \le k \le \lvert C \rvert$, exactly k distinct parties
`seed, rounds`	Master seed for keyed RNG streams; default CLI horizon
`[groups.<name>]`	voting, movement (stationary | random_walk | plugin), max_step $\in [0,1]$, communication (none | truthful | target_report), graph, proximity_radius $\in [0,\sqrt{2}]$, parameters
`capabilities`	15 permissions. Sensing, sight, interception, signal and roster default to false; send, receive, move and modify default to true
`[[people]]`	id, group, strict preference (or ideal $\in [0,1]^d$), position $\in [0,1]^2$, per-person overrides
`[network]`	radius; proximity_mode dynamic | frozen | static; directed persistent and campaign edges; private_channels
`[[interventions]]`	Scheduled per round and person: preference, position, group, one-round ballot, edge and channel edits
`[convergence]`	case = direct_best_response | source_local_dominance | certified_conformity (guard-checked)

Validation is canonical: every Scenario, including programmatic replacements, is re-parsed. Invalid input raises ElectionError before any state exists.

Configuration II: one table per mechanism, each with a budget

Table	Key fields
`[[coalitions]]`	members, target, report / sight / eavesdrop budgets per round and in total
`[campaign_controller]`	target, policy none | random | target_heuristic, lifetime and per-round budgets (1 per edge)
`[signal_controller]`	target, policy truthful | bounded_random | target_biased, schedule; cost = L1/2
`[nomination]`	m, eligible, mode endogenous | fixed | random
`[alignment.attack]`	target, targets, policy, per_person_round, total_budget (L2 path)
`[detector]`	target, threshold: alert if Δ target share > threshold
`[leaders]`	policy fixed | feedback, step, budget, signal_noise, perception_weight
`[majority_estimation]`	poll_counts, memory_rounds = 10, poll_weight = 0.5

```toml
# examples/all-enabled.toml
k = 2
seed = 2026

[campaign_controller]
target = "alice"
policy = "target_heuristic"
lifetime_budget = 1

[signal_controller]
target = "alice"
policy = "target_biased"
schedule = "repeated"

[alignment.attack]
target = "alice"
targets = ["bob"]
total_budget = 0.3

[detector]
target = "alice"
threshold = 0.1
```

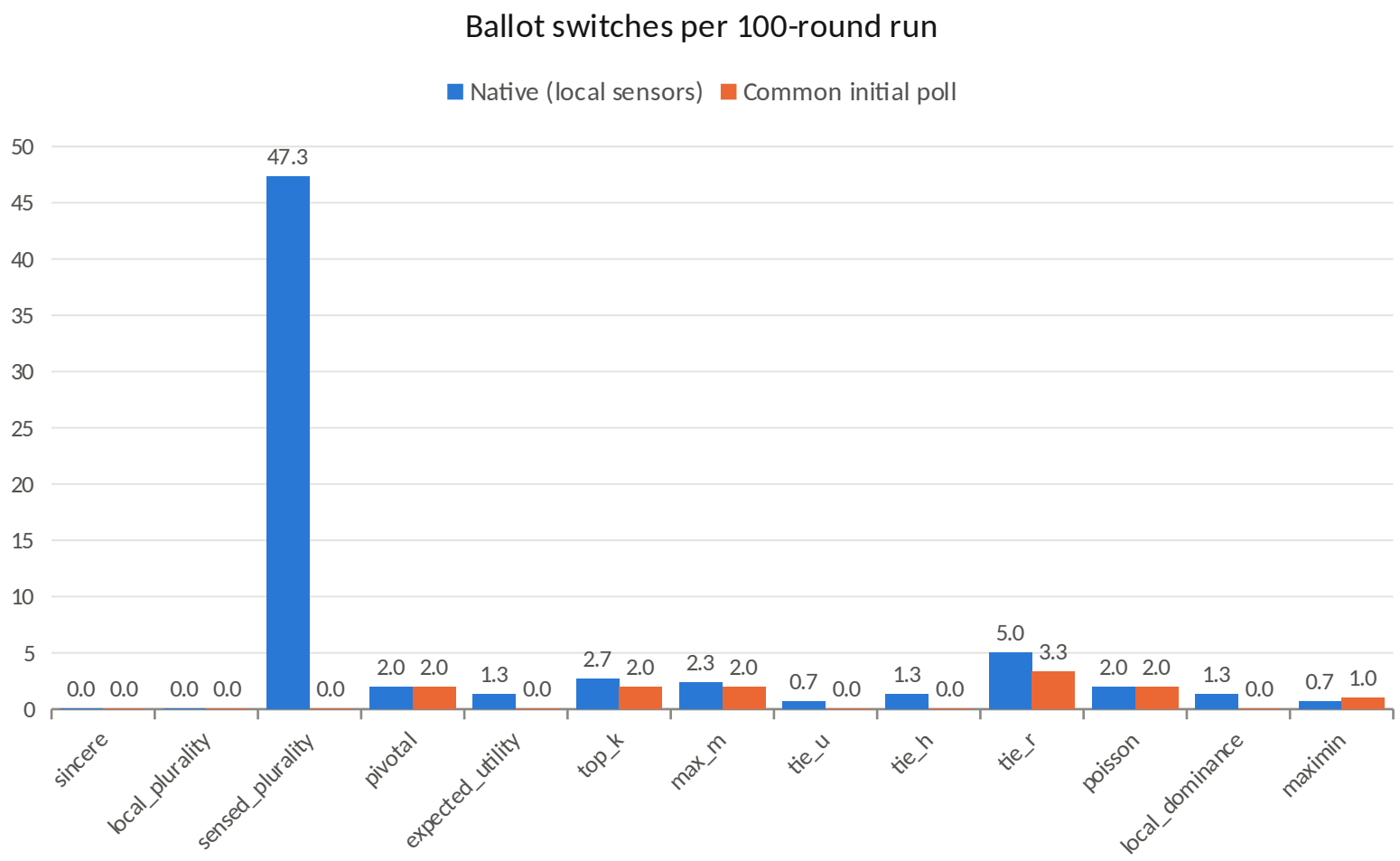
26 built-in voting rules, from sincere to certified

Controls (3)	sincere · local_plurality · sensed_plurality
Strategic (10)	pivotal · expected_utility · top_k · max_m · tie_u · tie_h · tie_r · poisson · local_dominance · maximin
Epistemic (6)	de re · dominant · pessimistic, each with a bounded-signal variant
Coalition & signal (4)	sight_plurality · overheard_plurality · coalition_fusion · signal_confident
Certified (3)	direct_best_response · source_local_dominance · certified_conformity

Strategic defaults: top_k = 2, samples = 200, uncertainty_radius = 2, max_states = 200,000, information_source native | poll. Plugins: python:module:function, hashed for replay but not sandboxed.

Study A: conformity churns; strategic rules mostly hold the sincere winner

9 voters, 3 parties, random walk, 100 rounds, seeds 101/202/303. Mean ballot switches per run.



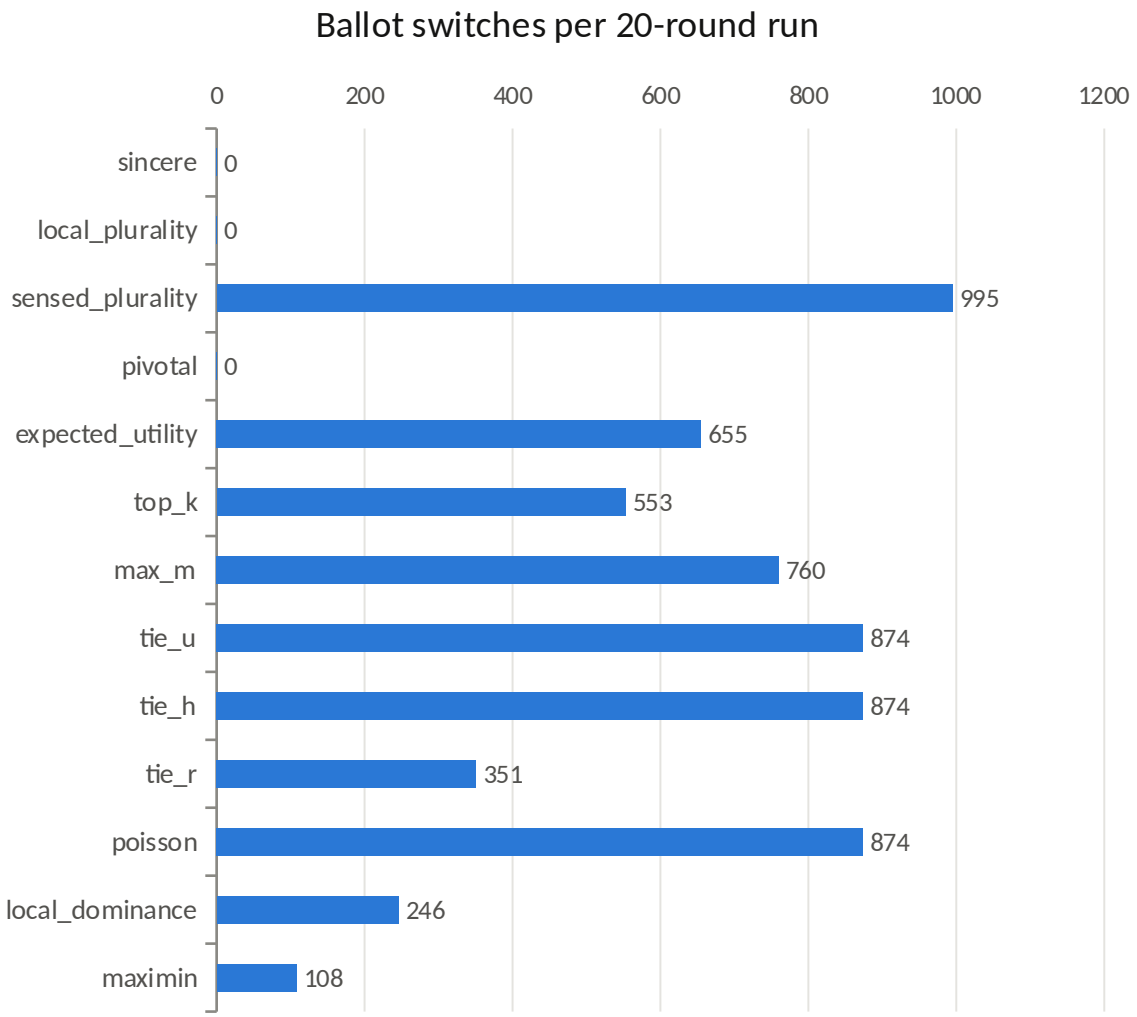
sensed_plurality: 95 switches on seed 101, hitting the horizon on 2 of 3 seeds

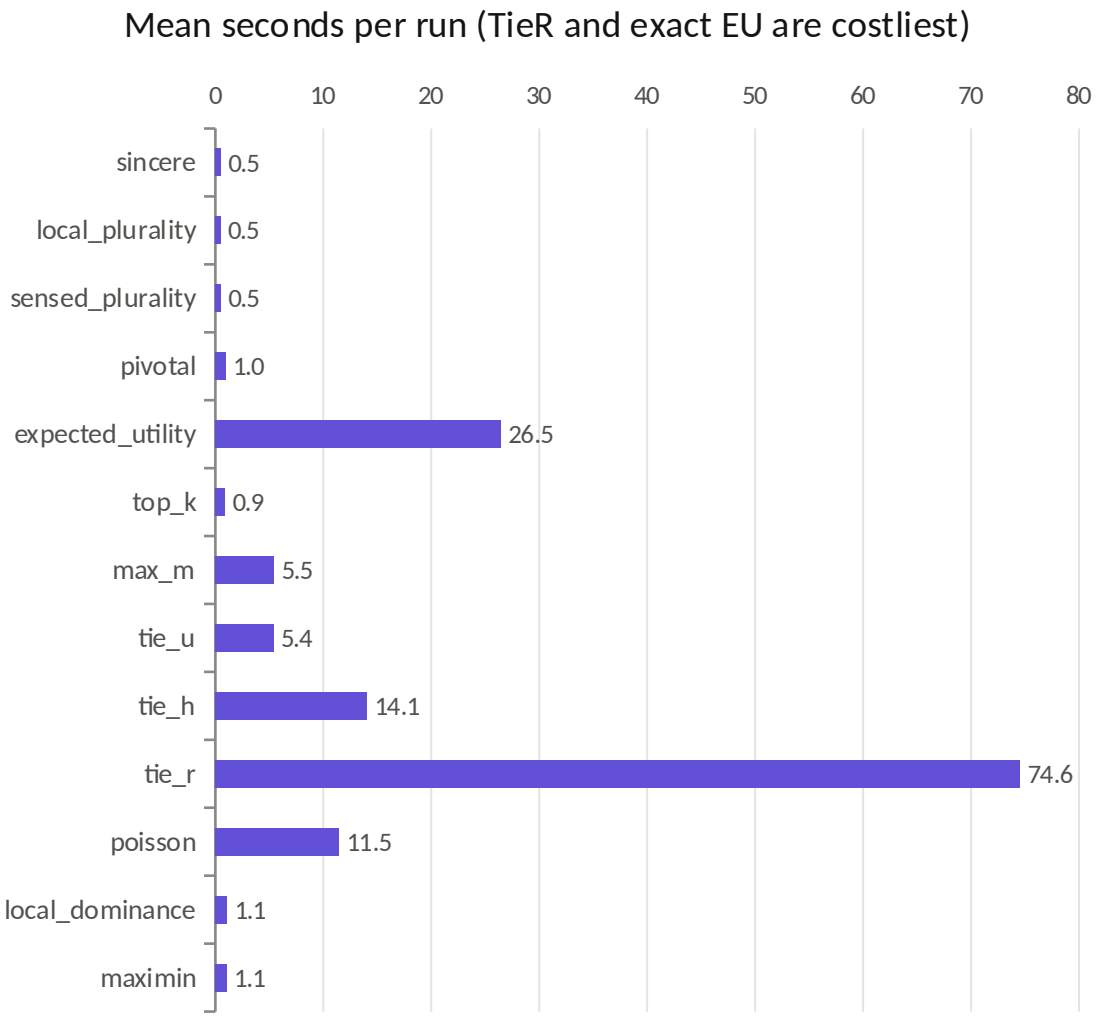
28 of 30 strategic runs keep sincere winner C; TieR and maximin each end one run on B

With a common poll, all 30 strategic runs stabilise at 0–4 switches, so the information source drives the differences

69 runs, 0 failures. Policies without a common-poll arm show 0.

Study A at 500 voters: cost and churn diverge sharply





No universal convergence guarantee, so five domains are certified

Counterexample

Two voters, u: A≻B and v: B≻A, both copy their only neighbour (sensed_plurality) on edges u⇄v.

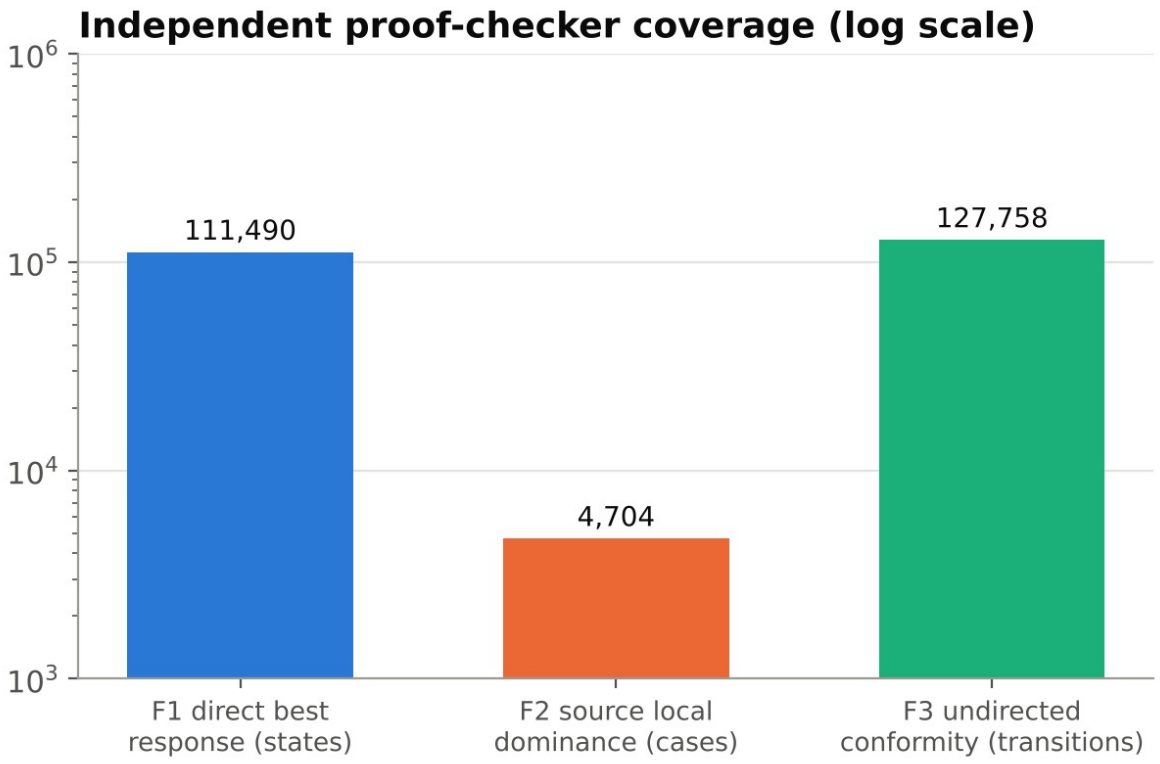
t	u	v	winner
0	A	B	A
1	B	A	A
2	A	B	A
…	…	…	A

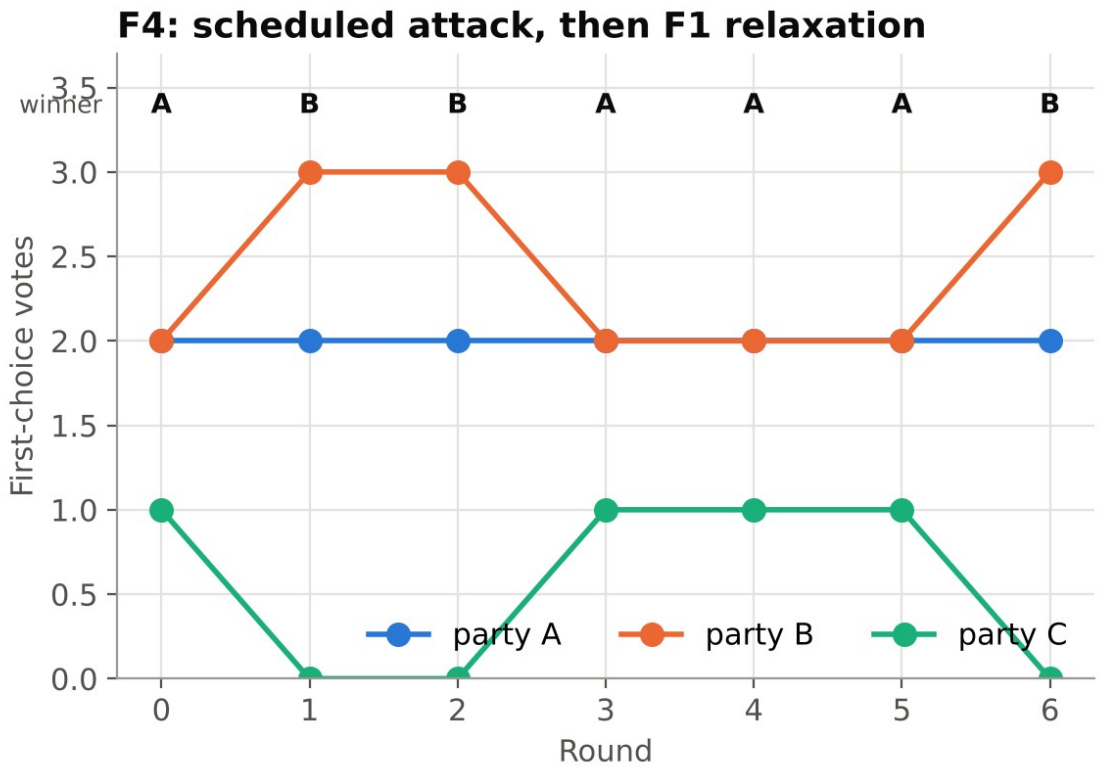
Ballots flip forever while the winner never changes. A stable winner is not convergence.

Case	Behaviour	Guarantee	Change bound
F1	Direct best response to truthful aggregate	Pure Nash equilibrium	$\le m^2N^2$ ($\le mN$ truthful)
F2	Homogeneous local dominance, L1 radius r	Voting equilibrium	$\le N(m-1)$
F3	Conformity on fixed undirected graph	Local fixed point	$\le m\|E\| + N(m-1)$
F4	Finite attacks, then F1 relaxation	Equilibrium of final election	F1 bound after last event
F5	Finite attacks, then F3 on same graph	Local fixed point	F3 bound after last event

F3 potential: $\Phi = m \cdot A(b) + \Sigma\, q_i(b_i)$ rises by at least 1 on every change. Sequential cyclic scheduler; unsupported settings fail certification explicitly.

Certified runs reach equilibrium, and the proofs are checked exhaustively





F1: certified-direct

C,B,A,A,B → voter 1 moves C→B. Nash equilibrium, B wins 3–2, 1 change (bound 15).

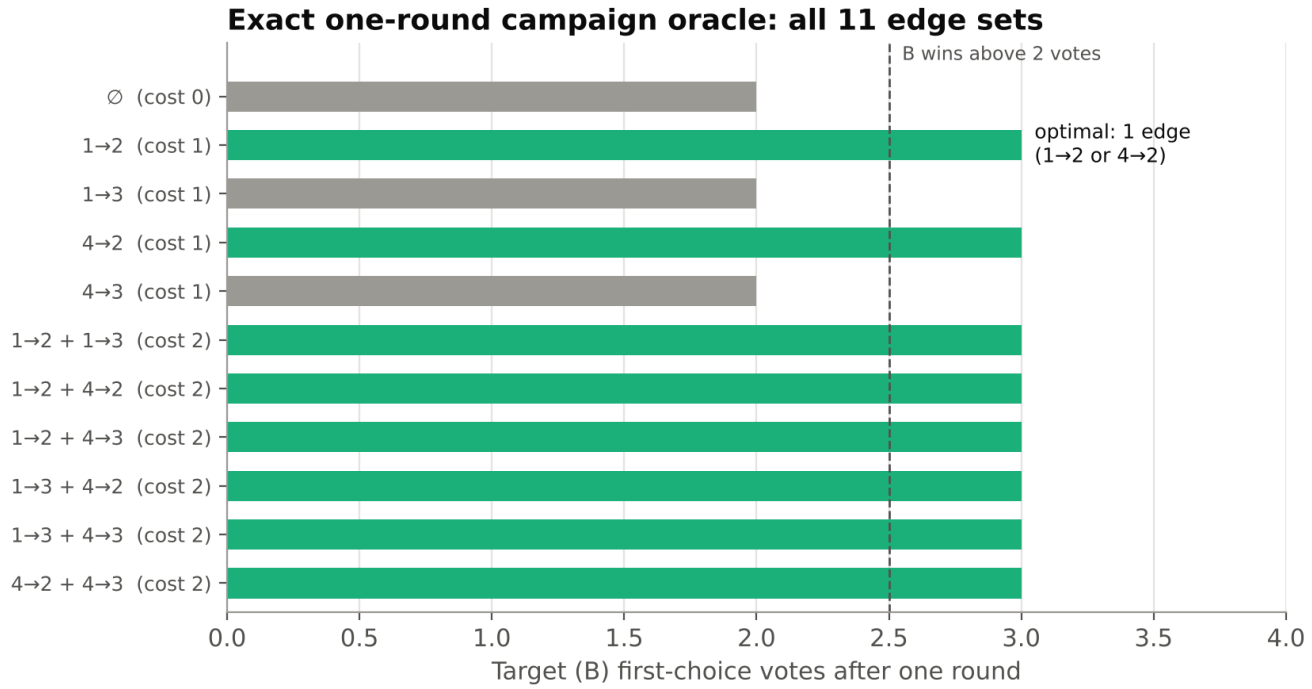
F3: certified-conformity

(A,B) → (B,B): a fixed point after 1 change. The simultaneous version would cycle.

F4: attack, then relax

An attack at round 3 resets C and A wins; F1 restores the B equilibrium at round 6.

Optimal solutions: exact oracles return the cheapest winning manipulation



Campaign optimum (budget 2)

All 11 eligible edge sets were enumerated. One edge, 1→2 or 4→2, is the cheapest win (B: 2 → 3 votes). The edges 1→3 and 4→3 alone are wasted.

Conditional epistemic equilibria

9 profiles and 72 evaluations give exactly two equilibria, {A,B} and {B,B}, reproducing the source paper's §5.2.

F1 / F2 equilibrium

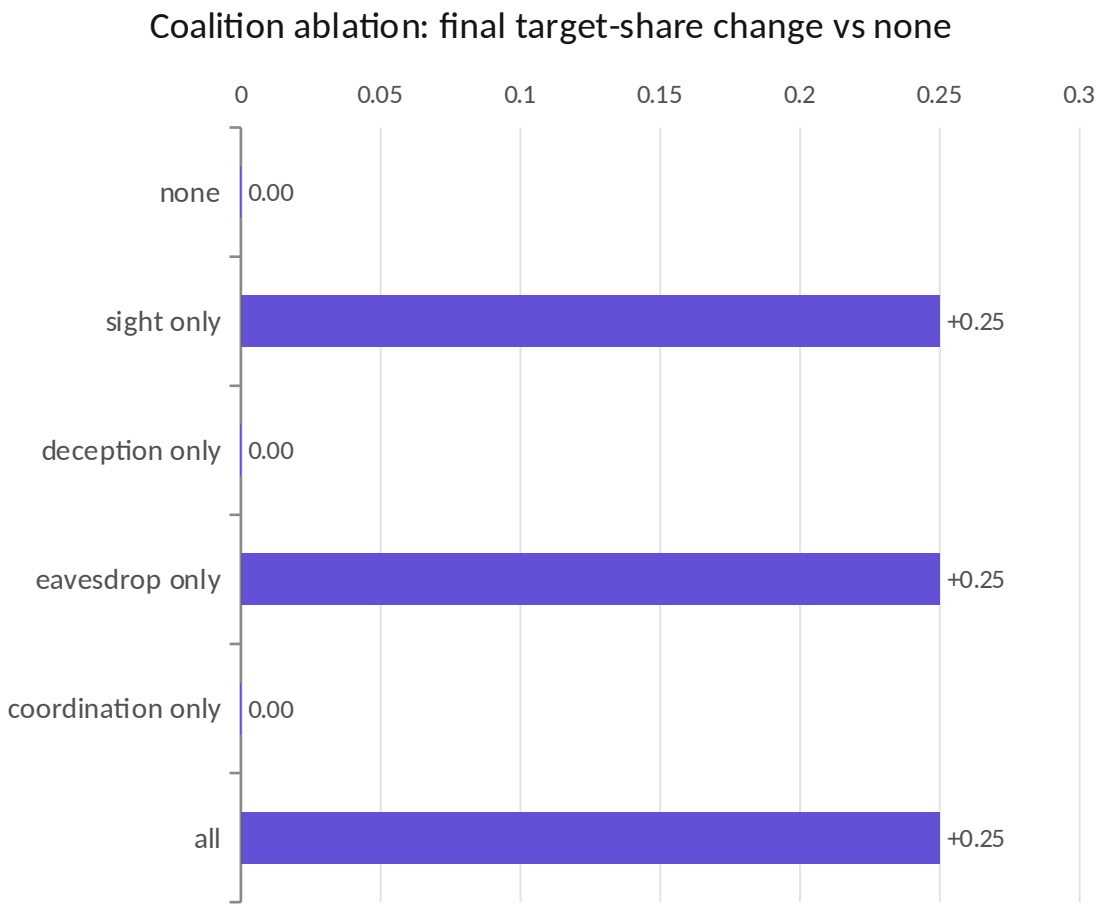
The compromise C→B is the best and dominance response, and it is reached in 1 step.

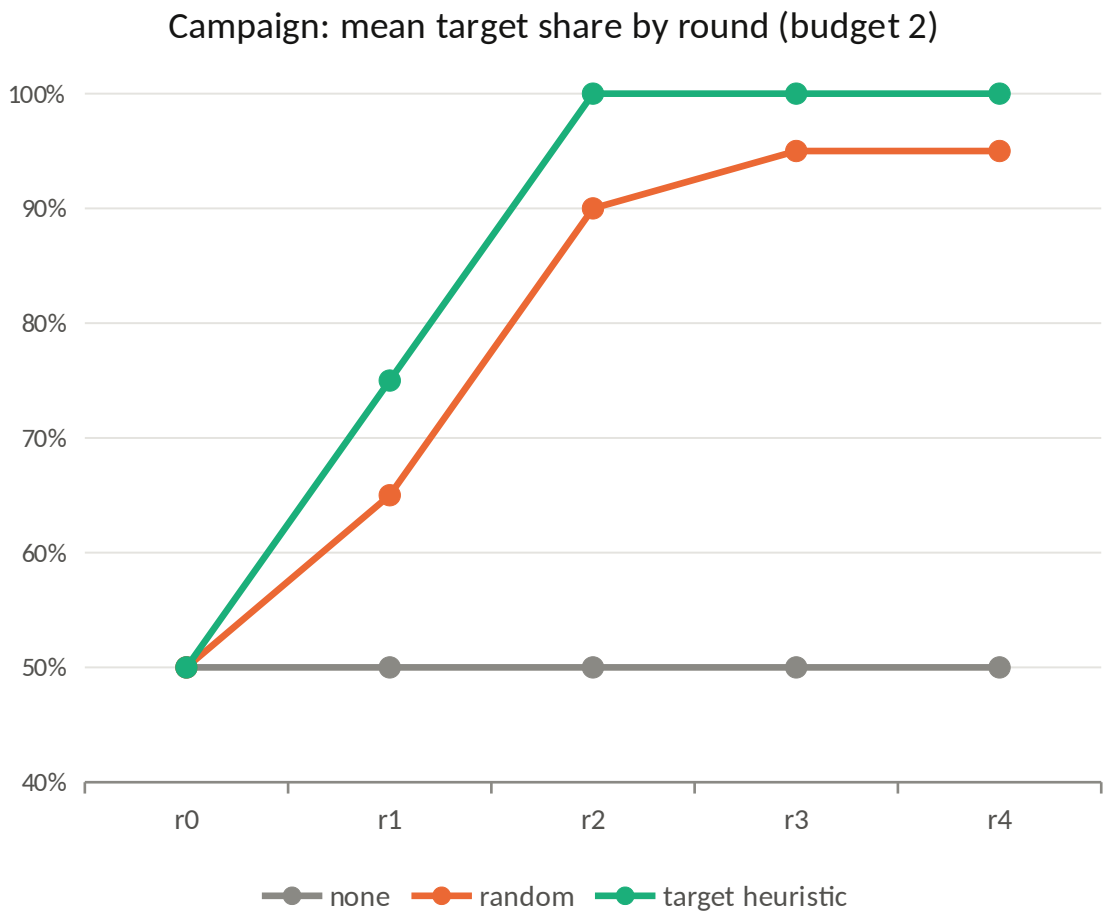
Seven paired mechanism studies: 5 seeds each, zero failures

The arms share scenario, seed and movement draws, and positions are verified equal. The examples have 3–4 voters, so these runs validate controls rather than estimate effect sizes.

Study	Arms	Outcome on example electorate	Verdict
B Proximity	dynamic vs frozen	Exposure L1 = 0 at 3 voters; ≈5k contact-rounds at 200 voters	Null
C Coalition	6 capability arms	Sight and eavesdrop flip to target in 5/5 (+0.25 share)	Effect
D Campaign	none / random / heuristic	Heuristic wins from round 1 in 5/5; random catches up later	Effect
E Public signal	3 responses × 3 polls	Budget spent, final share unchanged in all 45 runs	Null
F Nomination	endogenous / fixed / random	Fixed control is frame-identical; random adds bob in 4/5	Control OK
G Alignment	4 drift / attack arms	Targeted attack wins 5/5, random 3/5; detector recall 100% / 60%	Effect
H Leaders	fixed/adaptive × signal	Perception error ≤ 0.10, no winner change; invariant holds	Null

Coalitions and campaign edges: information and links flip the result



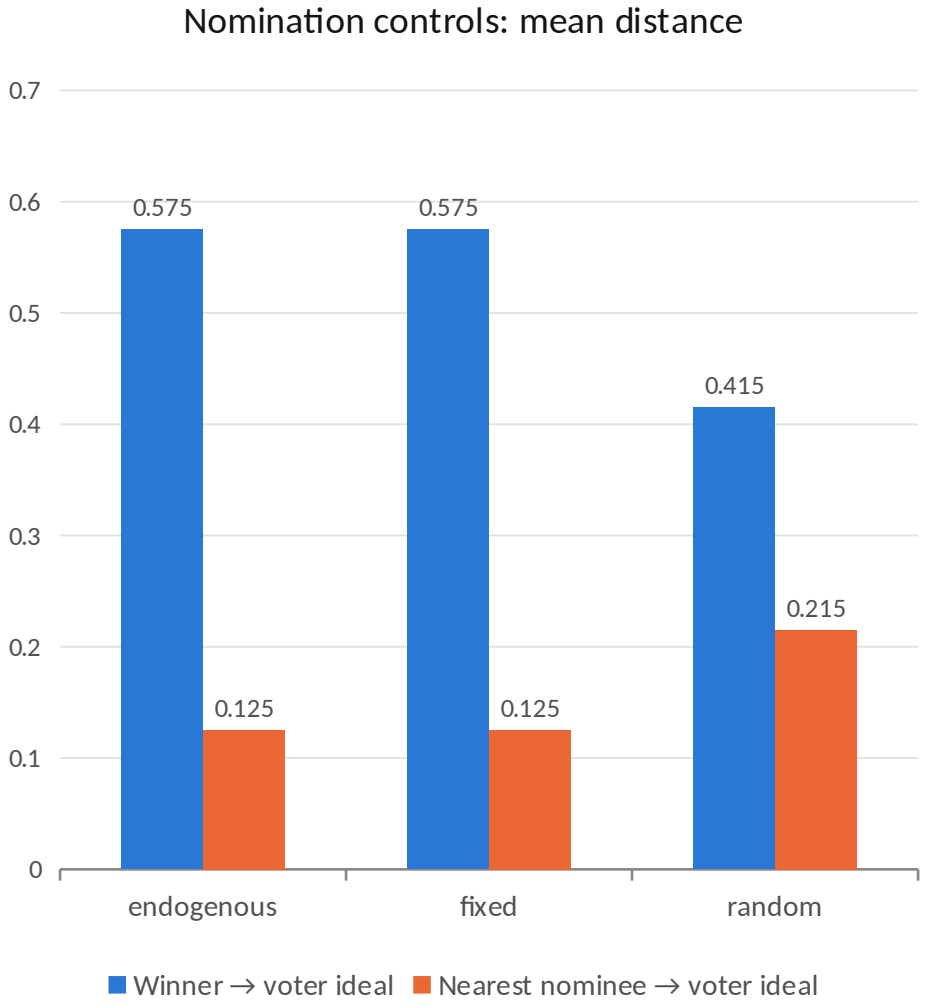


Deception alone and coordination alone do nothing here; "all" equals the best single capability. Both campaign controllers spend the full budget.

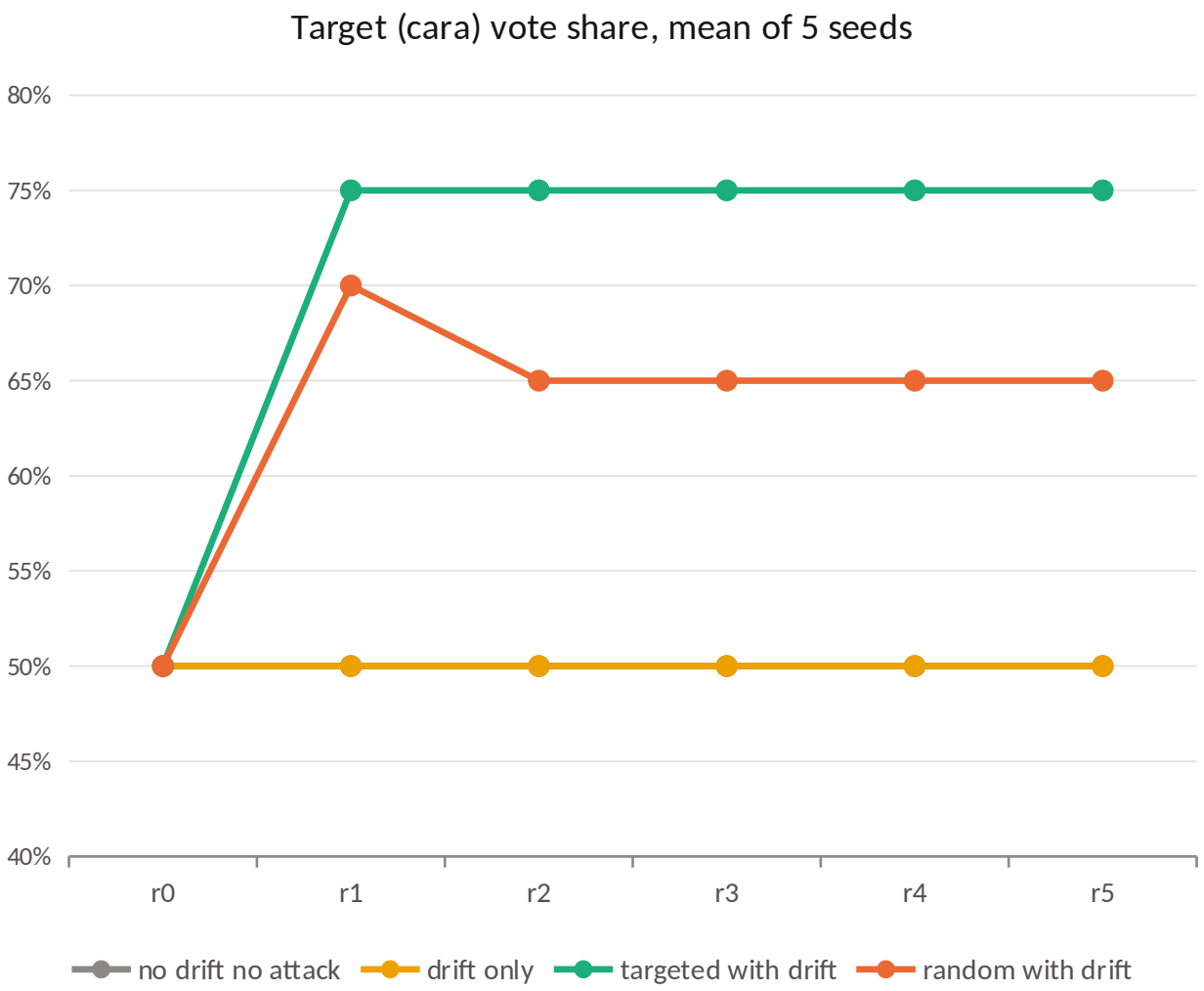
Public polls and nomination: bounded distortion is too weak here

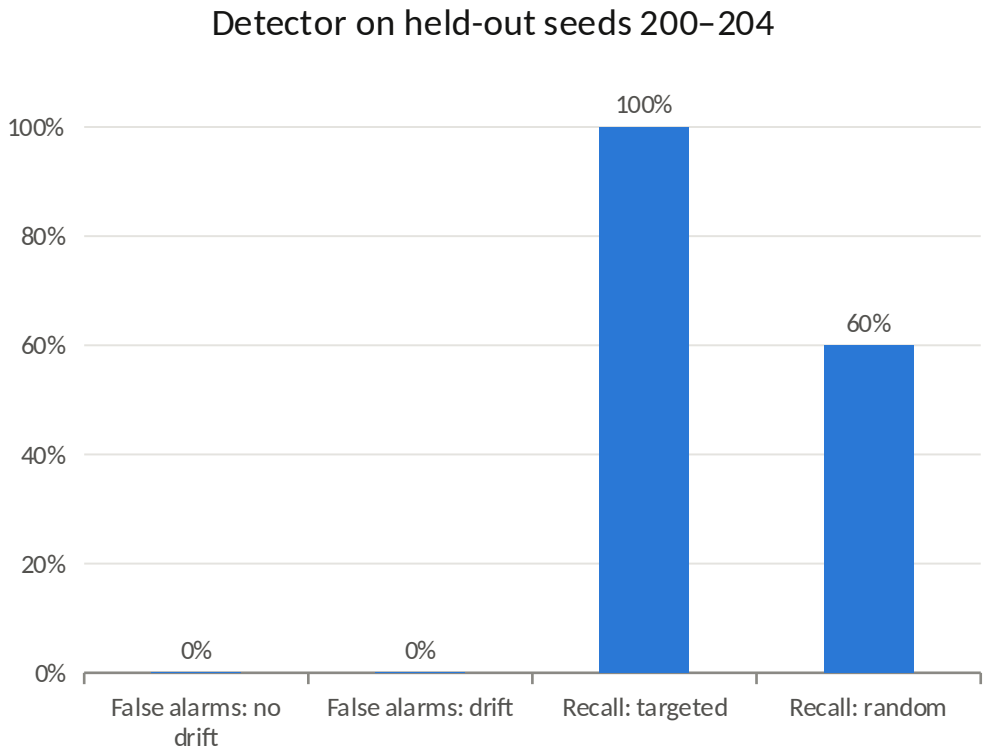
Response \ poll	truthful	bounded_random	target_biased
sincere	0 switches spend 0	0 switches spend 2	0 switches spend 2
signal_confident	0 switches spend 0	0 switches spend 2	5 switches spend 2
epistemic_signal_dominant	0 switches spend 0	2 switches spend 2	0 switches spend 2

3×3 factorial, 5 seeds per cell; switches are totals over the 5 seeds. Distorted polls spend their full budget of 2 and trusting or dominant voters occasionally switch, but target B's final share changes by 0.00 in every cell.



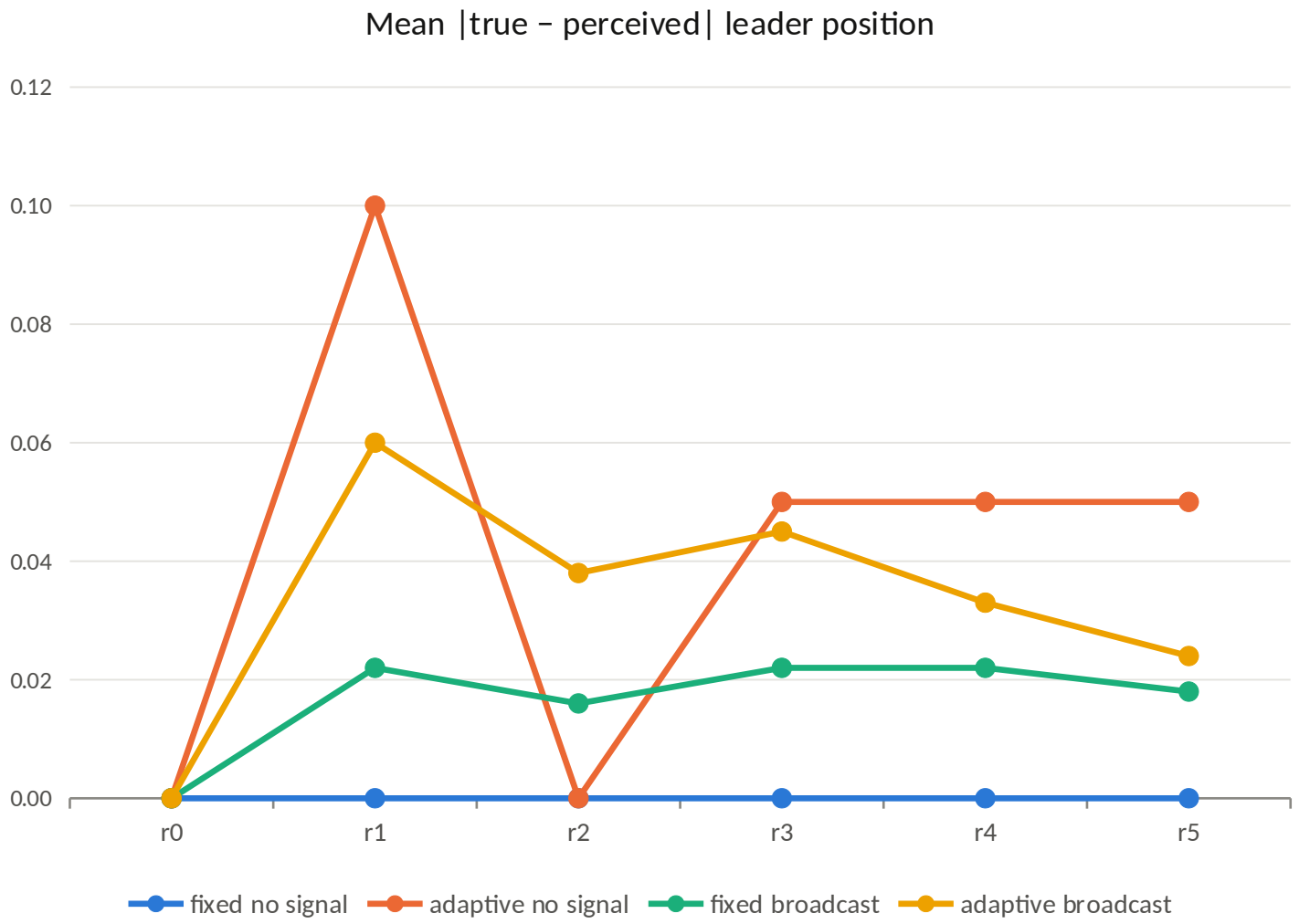
Alignment attacks win, and the aggregate-only detector catches them





The threshold is calibrated on disjoint no-attack seeds 100–104 (0.0). Every alert comes at onset (0-round delay), before the third consecutive win. Attack spend is 0.35 of L2 budget.

Hidden leader adaptation moves platforms, not outcomes

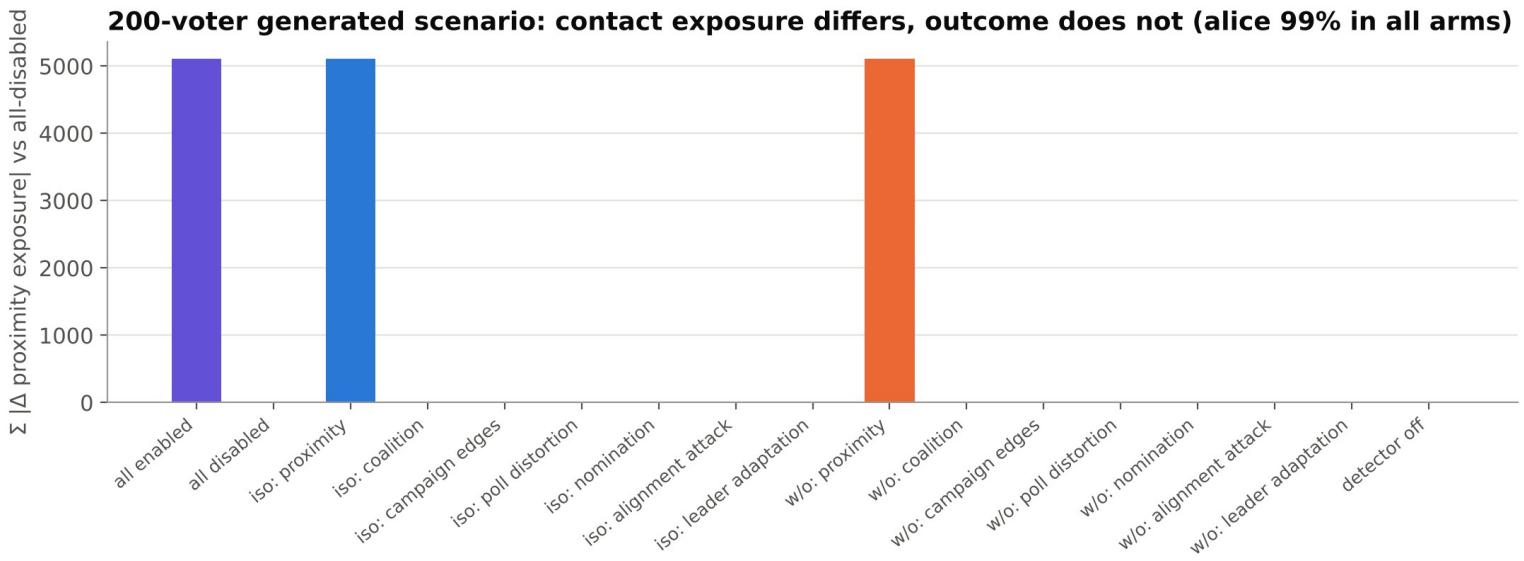


Feedback leader alice repeats a move after support rises and reverses after it falls. She spends her full 0.5 budget.

Without signals, perceptions freeze, so error peaks at 0.10. Noisy broadcasts (±0.05) cut it to about 0.06.

Winner unchanged in every round of all 20 runs. The no-signal invariant (all non-leader fields identical) holds 5/5.

Combined 17-arm matrix: valid infrastructure, but a ceiling effect



4-voter all-enabled

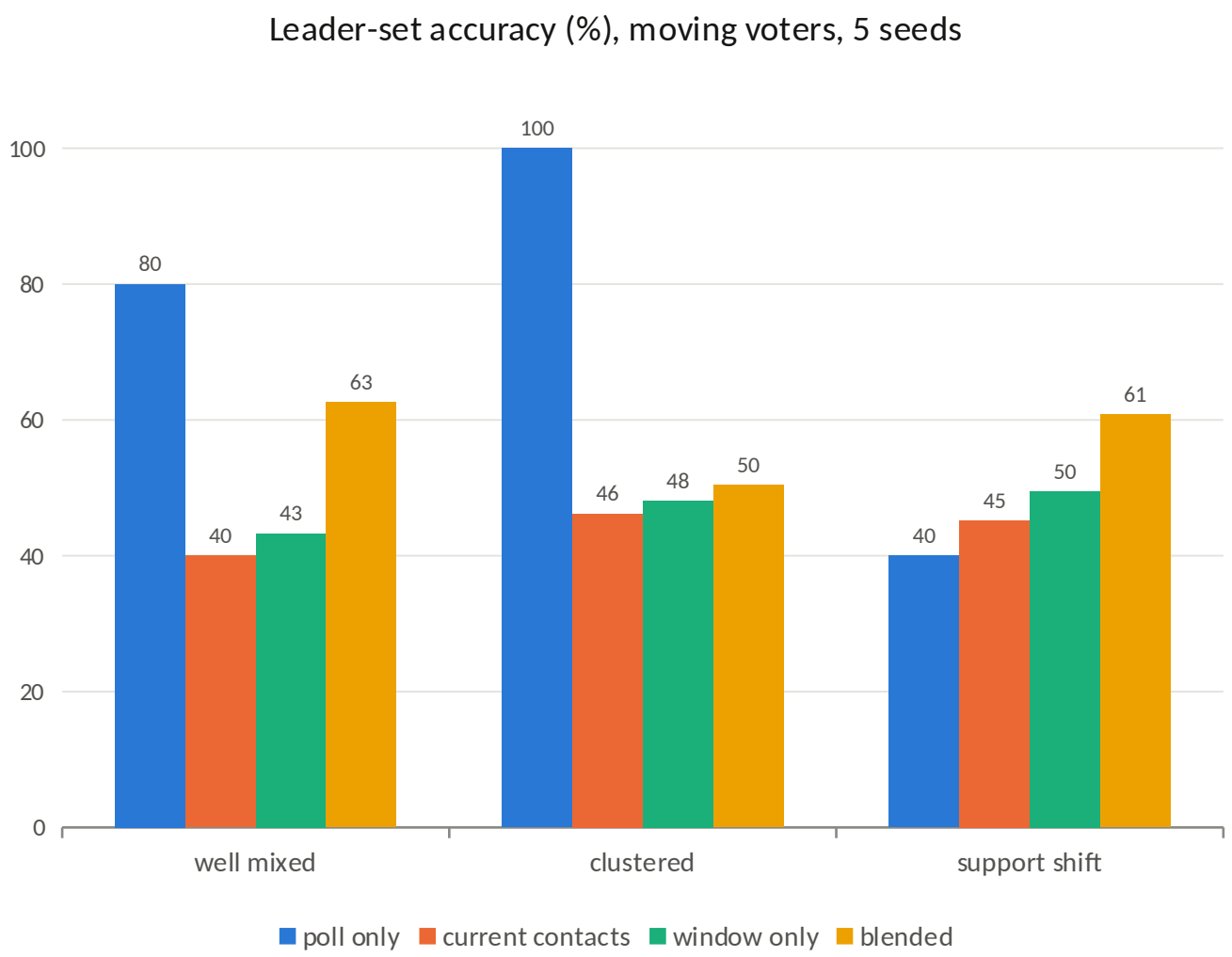
5 seeds × 17 arms, 0 failures. Alice wins all 6 rounds in every arm; nomination already gives her a majority.

200-voter generated

3 seeds × 17 arms, 0 failures. Alice holds 99% in every arm, and proximity changes exposure by ≈5k contact-rounds.

Arms: all enabled · all disabled · 7 isolated · 7 leave-one-out · detector-off, which checks that every non-detector field is identical. Next step: a balanced, contested electorate.

Majority estimation: local memory helps most when the poll goes stale



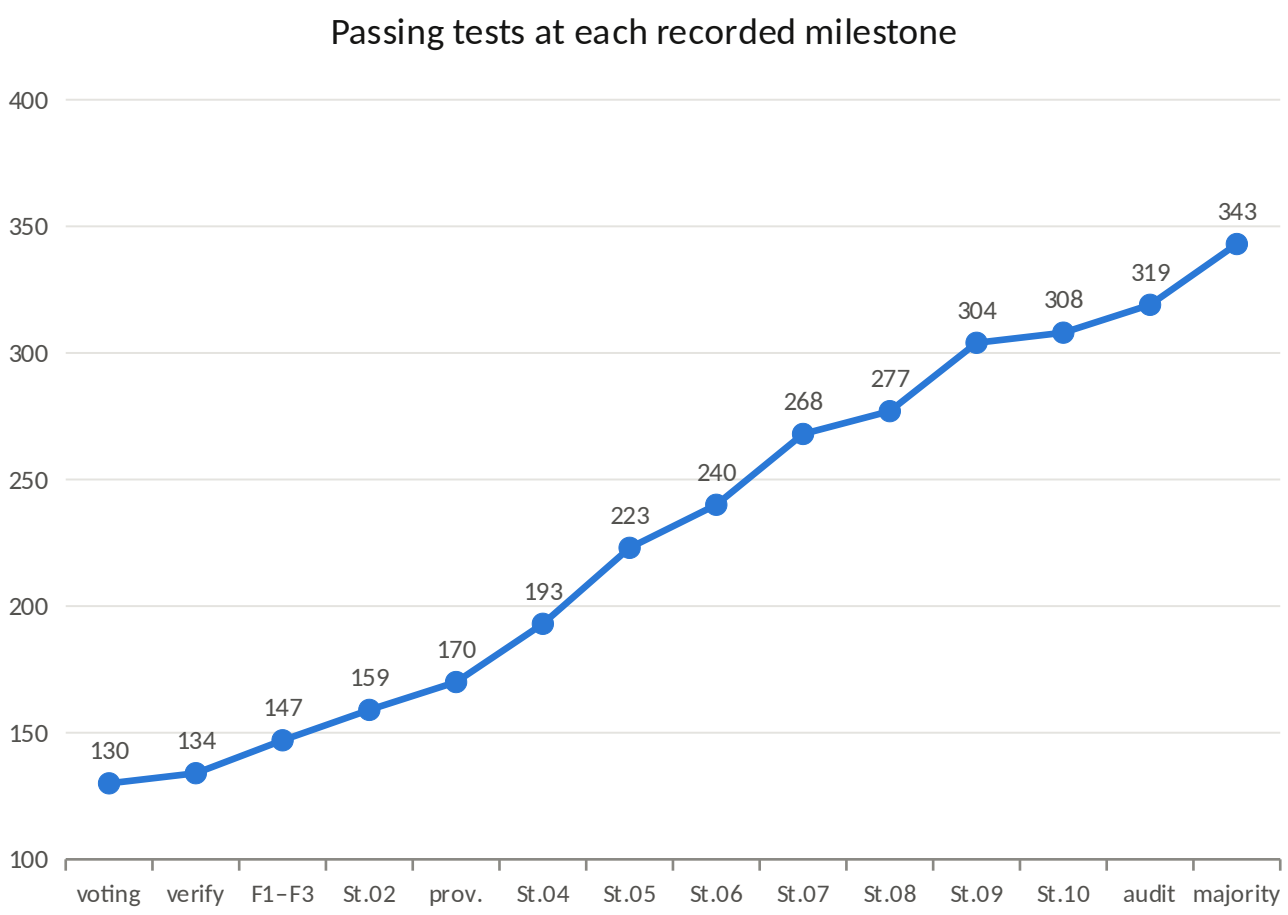
Passive belief: w·poll + (1−w)·recent neighbour share. It never changes ballots.

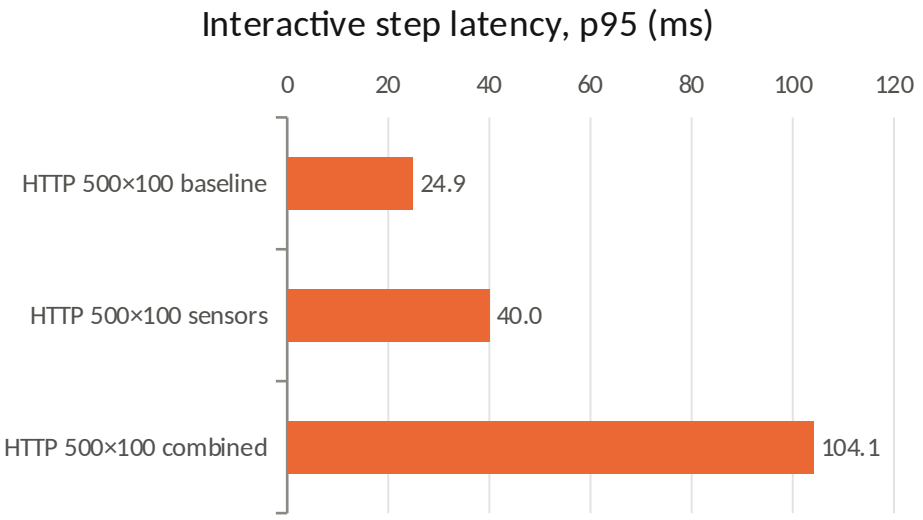
30 trajectories, 240 arms, 0 failures; the numbers match the repo's documented report exactly

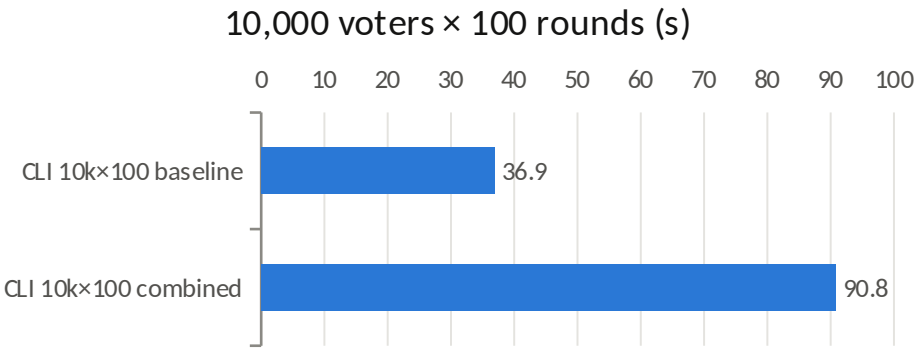
Support shift: after the round-50 shift the blend leaves 12.8% unrecovered, against 100% for the stale poll

Sensitivity: K=20 (error 0.078) and w=0.75 (0.046) each beat the defaults (0.085). Defaults unchanged.

Engineering quality: tests grew with every stage; scale is measured







Independent oracles: 80 randomised exact-EU cases, brute-force graphs, and exhaustive F1–F3 checks. All 31 examples replay. Benchmarks are single runs (i7-12700H), not guarantees.

Conclusions and next steps

1 **Built**

A reproducible, permission-scoped simulator with 7 mechanisms, 26 voting rules, replayable checkpoints and a browser UI. 343 tests pass.

2 **Proved**

Full-model convergence is refuted, but F1–F5 give certified equilibria. The exact oracles find the cheapest winning manipulations.

3 **Observed**

Information capabilities (sight, eavesdropping), campaign edges and alignment attacks flip small elections. Bounded poll distortion and leader adaptation do not.

4 **Next**

Freeze the Stage 11 protocol: balanced electorates, ≥30 held-out seeds, run-level uncertainty, a Top-2 parity fix, a rejected-action ledger and browser acceptance.

Report: docs/presentation_docs/Eclipse_Project_Report.pdf · commands to reproduce every figure: docs/presentation_docs/README.md